

Thermal Analysis of a Flip-chip Package

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Abstract—this document describes a study on the influence of changing parametric values on the thermal performance of a flip-chip package. The analytical processing and modeling of the package was done with Ansys Mechanical. The material study investigates the effect on thermal performance of different types of solder material, underfill material, lid material and TIM (Thermal Interface Material) with different thermal conductivities. The loading parametric study finds the effect of application of different uniform power values (heat flow), convection coefficient values and radiation emissivity values. The last study conducted is about the effect of different geometric parameters on the thermal performance of a flip-chip package. Namely, the effect of different values of lid thickness, TIM thickness, substrate core thickness and bulk silicon thickness are investigated. The results of different models are documented, and conclusions are made.

Keywords—flip-chip; package; thermal

I. INTRODUCTION

Temperature management in electronics packaging is crucial for optimal performance and extending the life of electronic components. This study evaluates the thermal performance of a flip-chip package where heat is produced by the active region (transistor layer). The heat is transferred from the active region through the underfill, bumps and the substrate at the bottom of the package and through the bulk silicon, TIM and metal lid at the top. A study of various parameters to compare different geometries and materials will be done to determine the optimal setup for this package. The parameters evaluated in the study are:

- 1) Maximum temperature in a flip-chip package
- 2) Convection heat dissipation path on the lid and substrate
- 3) Radiation heat dissipation on the lid
- 4) Temperature through-the-thickness of the package

The thermal performance of the package is evaluated through the study based on the values of given parameters depending on the type of material used, loading and geometry of the package.

II. PACKAGE DESCRIPTION

The flip-chip package was modeled in Ansys Mechanical including all material properties. Fig. 1 shows cross-section of the flip-chip package.

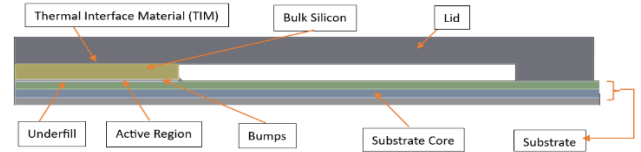


Figure 1. Cross-section of a flip-chip package

A semiconductor chip is positioned onto the top side of the substrate. It is attached to the substrate with solder bumps. The space between the substrate and the chip is filled with an underfill material for protection against environmental conditions and for adding mechanical strength. The chip consists of an active region (the transistor layer) and bulk silicon, containing different interconnections. A metal lid is attached to the top side of the chip. Between the metal lid and the back of the chip there is a layer of high conductivity TIM.

III. DEMO MODEL

The default (demo) model of a flip-chip package given has the following parameters:

- Solder type: SAC305
- Lid material: copper
- TIM thermal conductivity 3.6 W/m-K
- Uniform Power (Heat flow): 10W
- Convection coefficient value: 100 W/(m² K)
- Radiation emissivity 0.2
- Lid thickness: 1.25mm
- TIM thickness: 50µm
- Substrate core thickness: 0.41mm
- Bulk silicon thickness: 0.7mm.

If it is not specifically indicated that a certain value or a parameter is changed, then it is assumed to be set to the default value noted above.

The temperature performance of the default flip-chip package is described by the following temperature values:

- Maximum temperature: 48.874°C
- Convection heat dissipation path on the lid (out of 10 W generated by the active region) 9.7613 W
- Convection heat dissipation path on the substrate (out of 10 W generated by the active region) 0.23873 W
- radiation heat dissipation on the lid 1.1374e-007 W

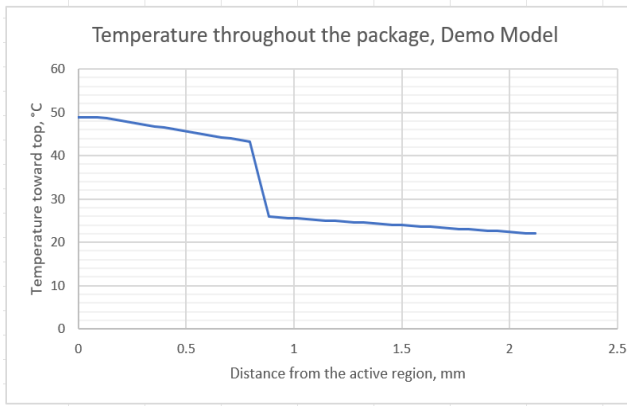


Figure 2. Temperature toward top of the package

Fig. 2 demonstrates the temperature throughout from the active region to the top of the package. Fig. 3 demonstrates temperature throughout from the active region to the bottom of the package

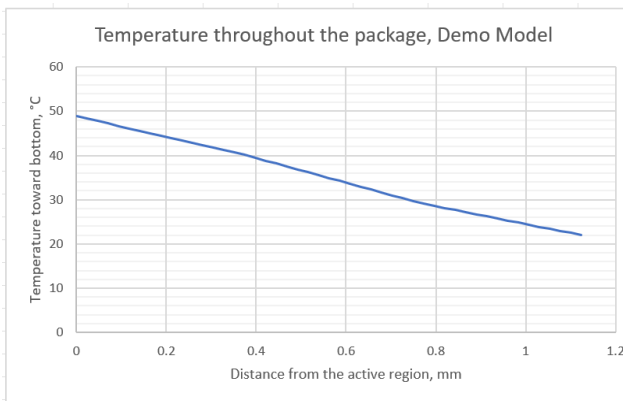


Figure 3. Temperature toward bottom of the package

IV. MATERIAL PARAMETRIC STUDY

A. Solder Type SAC305 vs SnCu.

During the study for solder type, the thermal performance of SAC305 is compared with SnCu thermal performance. Table 1 demonstrates the results.

Table 1. Thermal performance for solder material SAC305 vs SnCu

	Solder Type	
	SAC305 Demo	SnCu
Maximum temperature, °C	48.876	48.876
Convection Heat dissipation path on the lid, W	-9.7639	-9.7633
Convection Heat dissipation path on the substrate, W	-0.23614	-0.23675
Radiation heat dissipation on the lid, W	-1.1377e-007	-1.1377e-007

After comparing the numerical values of solder type SAC305 and SnCu from Table 1, it ~~is obvious~~ can be seen that there is ~~very little~~ difference in thermal performance between them. The maximum temperature of the package remains the same in both cases. That is explained by very close values of thermal conductivity for both of them. SAC305 has 0.05 W/mm·K and SnCu has 0.066 W/mm·K. Thermal conductivity is the ability of a material to conduct heat. The very minor difference between these two materials is that in case of SnCu a little more heat is dissipated through the substrate due to its little higher thermal conductivity. The negative sign means heat loss. Overall temperature graphs throughout the package were overlapping toward top and bottom of the package meaning that there is no considerable difference for the thermal performance of a flip-chip package no matter what type of the solder material is chosen.

B. Underfill Material: Underfill 1 vs Underfill 2

The study of the Underfill 1 vs Underfill 2 materials demonstrated results similar to the results of solder type materials. The temperature performance of the package in both cases was practically identical with a very minor difference of heat dissipation toward the bottom of the package. It is also explained by very close thermal conductivity of Underfill1 and Underfill2, which are 0.0005 W/mm·K and 0.0004 W/mm·K accordingly.

C. Lid Material: Copper vs Stainless Steel vs Aluminum

This study demonstrated significant difference between the thermal performances of the package depending on what material was chosen for the lid. Numerical results for different lid materials are presented in Table 2.

Table 2. Thermal performance of the package with different lid materials

	Lid Material		
	Copper Demo	Stainless Steel	Aluminium
Maximum temperature, °C	48.874	134.74	51.362
Convection Heat dissipation path on the lid, W	-9.7613	-9.0288	-9.7408
Convection Heat dissipation path on the substrate, W	-0.23873	-0.97117	-0.25921
Radiation heat dissipation on the lid, W	-1.1374e-007	-1.0494e-007	-1.135e-007

From this data it can be seen that the best performing material for the lid is copper. Coppers thermal conductivity is 0.39 W/mm·°C, aluminums is 0.237 W/mm·K and stainless steels is 0.0151 W/mm·K. Stainless steels thermal conductivity is considerably lower than copper and aluminum. That is the reason that the maximum temperature in the active region reaches 134.74°C.

Fig. 4 and Fig. 5 demonstrate similar picture by the temperature throughout the package toward top and the bottom of the package. While copper and aluminum have closer results in acceptable range, the disadvantage of using stainless steel is obvious.

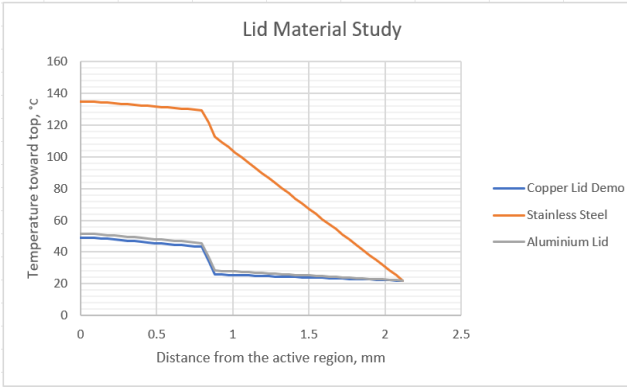


Figure 4. Temperature toward top of the package (Lid material study)

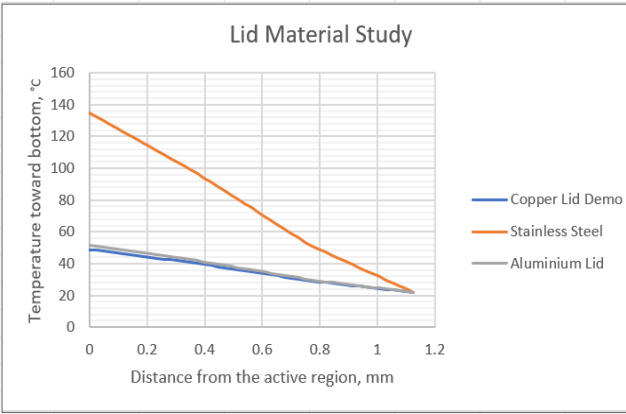


Figure 5. Temperature toward bottom of the package (Lid material study)

D. TIM Thermal Conductivity 0.1 vs 0.3 vs 0.5 W/m·K

The results of TIM thermal conductivity study are reflected below in Table 3.

Table3. Thermal performance for TIM

	TIM Thermal Conductivity		
	0.1W/m·K	0.3W/m·K	0.5W/m·K
Maximum temperature, °C	424.94	206.29	143.8
Convection Heat dissipation path on the lid, W	-6.4715	-8.3793	-8.9256
Convection Heat dissipation path on the substrate, W	-3.5285	-1.6207	-1.0744
Radiation heat dissipation on the lid, W	-7.4162e-008	-9.7114e-008	-1.0369e-007

The values of thermal conductivity for TIMs given for study are significantly less than the default value (3.6 W/m·K) hence the significant difference for the results. Though the overall temperature performance of the package is significantly different depending on what value is chosen for TIM thermal conductivity, none of them gives the results within the acceptable range. The lowest temperature achieved in the active region is 143.8°C that is still significantly higher than the default value and stressful for the chip. The temperature performance of the package is demonstrated in Fig. 6 and Fig. 7.

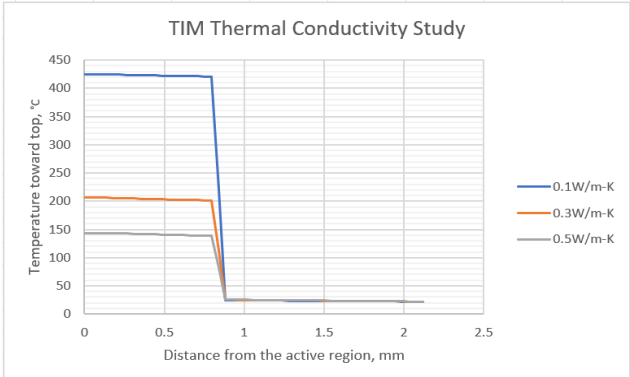


Figure 6. Temperature toward the top of the package (TIM conductivity study)

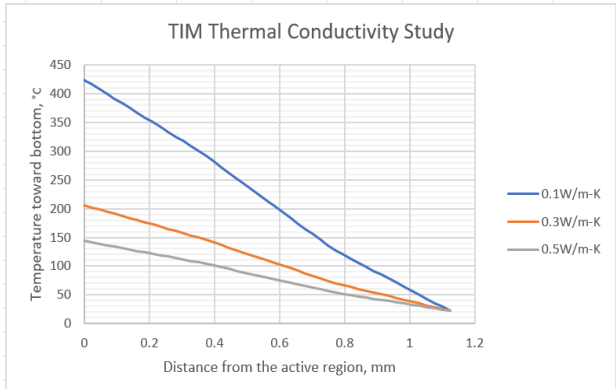


Figure 7. Temperature toward the bottom of the package (TIM conductivity)

V. LOADING PARAMETRIC STUDY

The loading parametric study will see the effects of different levels of power generated by the active region on thermal performance of a flip-chip package, the convection coefficient of the lid and substrate and radiation emissivity of the lid.

A. Uniform Power (Heat Flow): 5W vs 10W vs 20W

This study will evaluate the effect of different amount of heat generated by the active region of a flip-chip package on the temperature inside it. Table 4 demonstrates the numerical values acquired during the simulation.

Table 4. Thermal performance for different heat flow

	Uniform Power (Heat Flow)		
	5W	10W Demo	20W
Maximum temperature, °C	35.437	48.874	75.747
Convection heat dissipation path on the lid, W	-4.8806	-9.7613	-19.523
Convection heat dissipation path on the substrate, W	-0.11936	-0.23873	-0.47745
Radiation heat dissipation on the lid, W	-5.6869e-008	-1.1374e-007	-2.275e-007

After comparing the values of different parameters from the table, it is obvious that the more power (heat flow) is generated by the active region, the higher is the temperature in a flip-chip package. It is explained by the fact that the more powerful active region generates more energy (heat) that has to be dissipated by the package. The same tendency is demonstrated by the temperature throughout toward the top and the bottom of a flip-chip package in Fig. 8 and Fig. 9.

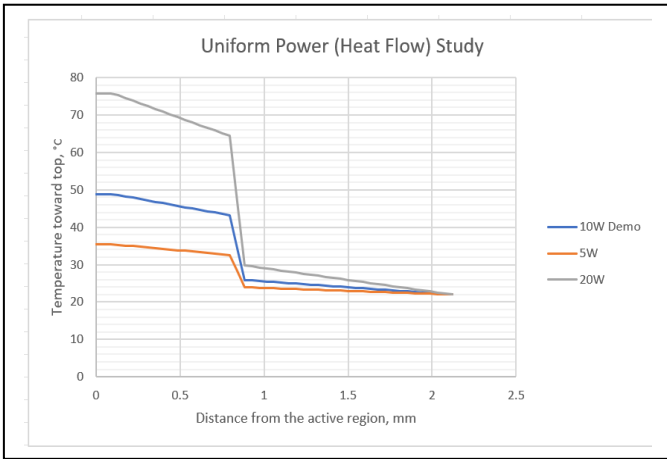


Figure 8. Temperature toward top of the package (Heat Flow study)

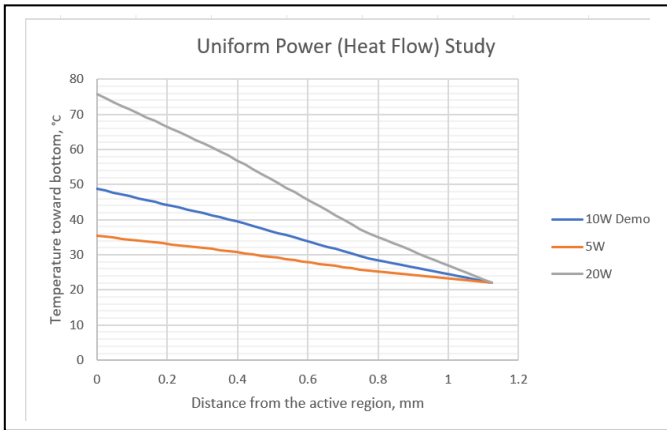


Figure 9. Temperature toward bottom of the package (Heat Flow study)

B. Convection coefficient value: 10 vs 100 vs 1000 W/(m² K)

Convection is the heat transfer from one place to another due to the movement of fluid. The convection coefficient is the rate of heat transfer between a solid surface and a fluid per unit surface area per unit temperature difference [1]. In our study, we compare three values which differ ten times from each other. The table 5 below shows the acquired results.

Table 5. Thermal performance for different convection coefficient values of the lid and substrate

	Convection coefficient value		
	10W/(m ² K)	100W/(m ² K) Demo	1000W/(m ² K)
Maximum temperature, °C	48.98	48.874	48.863
Convection heat dissipation path on the lid, W	-9.7605	-9.7613	-9.7614
Convection heat dissipation path on the substrate, W	-0.23954	-0.23873	-0.23864
Radiation heat dissipation on the lid, W	-1.1379e-006	-1.1374e-007	-1.1374e-008

After evaluating the numerical values from the table 5, it is noticeable that the bigger convection coefficient helps better convection of the heat through the package. In particular, more convective heat is transferred through the lid vs substrate of the package. Overall, the differences are very minor and the temperature performance of the package is not much affected. The graphs for the temperature toward top and the bottom of the package almost fully repeat each other and thus are not demonstrated here. The maximum temperature in a flip-chip package differs very insignificantly.

C. Radiation Emissivity Study: 0.2 vs 0.4 vs 0.6

Radiation is the heat transfer through the emission of electromagnetic waves [2]. Radiation emissivity is the measure of an object's ability to emit infrared energy. Emissivity can have a value from 0 (shiny mirror) to 1.0 (blackbody) [3]. In our study we change the emissivity of the lid to three different values and compare the results, which is reflected in Table 6 below.

Table 6. Thermal performance for different lid emissivity values

	Radiation Emissivity		
	0.2 Demo	0.4	0.6
Maximum temperature, °C	48.874	48.874	48.874
Convection heat dissipation path on the lid, W	-9.7613	-9.7613	-9.7613
Convection Heat dissipation path on the substrate, W	-0.23873	-0.23873	-0.23873
Radiation heat dissipation on the lid, W	-1.1374e-007	-2.2748e-007	-3.4122e-007

After analyzing data from Table 6, it is obvious that different radiation emissivity of the lid doesn't affect the thermal performance of a flip-chip package. Maximum temperature stays the same in all three cases. The same happens with the temperature throughout the package toward top and bottom. (Graphs are not demonstrated). The only difference is seen in the amount of heat radiation on the lead, that can be explained by different radiation emissivity of the lid. Still these values are too insignificant to be considered.

VI. GEOMETRY PARAMETRIC STUDY

The geometry parametric study will show how changing geometric properties of the flip-chip package will change its thermal performance. The geometries that will be changed in this study are the lid, TIM, substrate core and bulk silicon thickness.

A. Lid Thickness: 1.25mm vs 1.5mm vs 2mm

The lid thickness study compares the three lid thicknesses seen in Table 7. From this data it is seen that the thinner lid performs slightly better by having a lower maximum temperature.

Table 7. Thermal performance for different lid thickness

	Lid Thickness		
	1.25mm	1.5mm	2mm
Maximum temperature, °C	48.874	49.628	51.104
Convection heat dissipation path on the lid, W	-9.7613	-9.7555	-9.7443
Convection heat dissipation path on the substrate, W	-0.23873	-0.24454	-0.25572
Radiation heat dissipation on the lid, W	-1.1374e-007	-1.1366e-007	-1.1351e-007

In addition to the data presented in Table 7, Fig. 10 and Fig. 11 graph the temperature of the lid toward the top and bottom. These graphs confirm that a lid thickness of 1.25mm performs the best by maintaining a lower temperature when comparing to lid thicknesses of 1.5mm and 2mm.

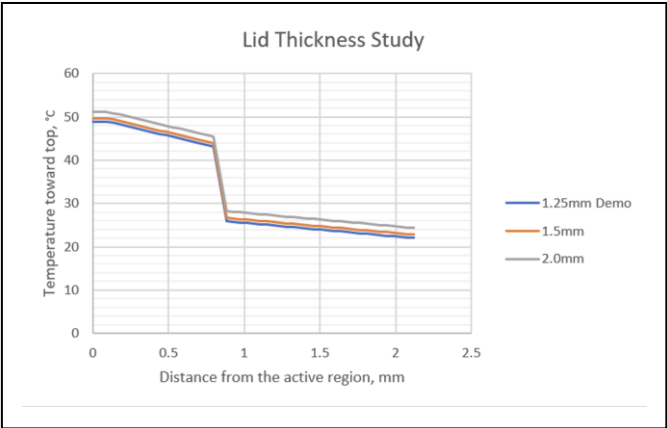


Figure 10. Temperature toward top of the package (Lid Thickness study)

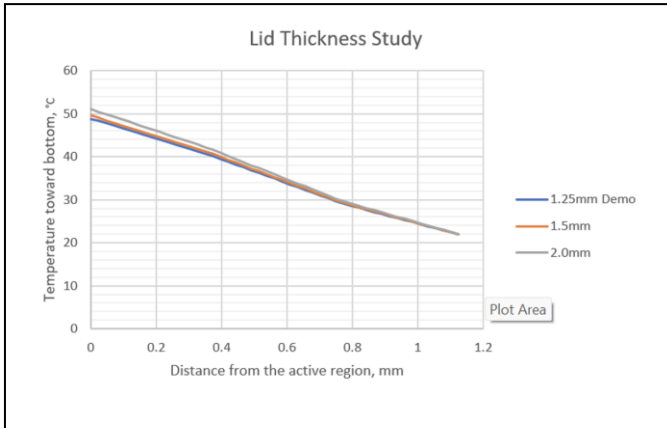


Figure 11. Temperature toward bottom of the package (Lid Thickness study)

B. TIM Thickness: 50µm vs 75µm vs 100µm

Comparisons between thermal performance of different TIM layer thicknesses was conducted with results presented in Table 8. TIM thickness affects the resistance to heat flow [4]. These results show that the thinnest TIM layer with the thickness of 50µm helps the package maintain a lower maximum temperature of 48.874 °C.

Table 8. Thermal performance for different TIM thickness

	TIM Thickness		
	50µm	75µm	100µm
Maximum temperature, °C	48.874	57.185	65.382
Convection heat dissipation path on the lid, W	-9.7613	-9.6873	-9.6146
Convection heat dissipation path on the substrate, W	-0.23873	-0.31273	-0.38541

Radiation heat dissipation on the lid, W	-1.1374e-007	-1.1285e-007	-1.1197e-007
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Similar results can be seen in Fig. 12 and Fig. 13 by plotting the temperature towards the top and bottom of the package. These graphs show that the package maintains the lowest temperature using a thickness of 50µm.

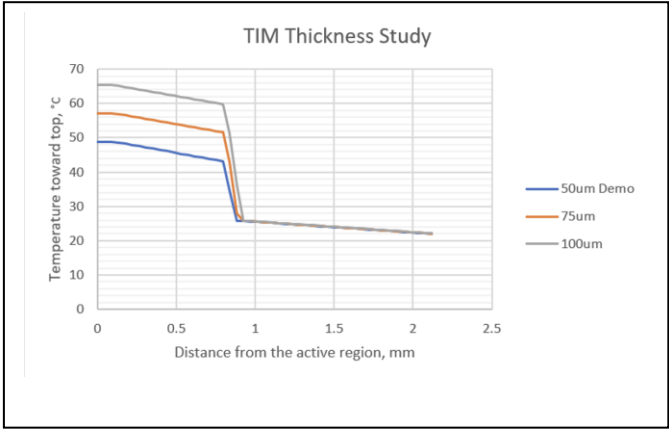


Figure 12. Temperature toward top of the package (TIM Thickness study)

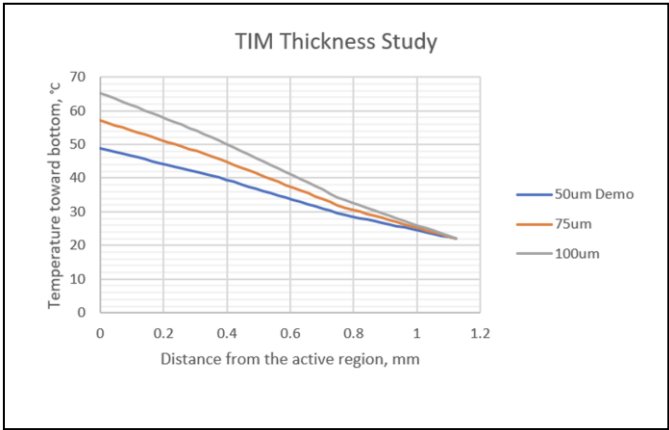


Figure 13. Temperature toward bottom of the package (TIM Thickness study)

C. Substrate Core Thickness: 0.41mm vs 0.5mm vs 0.6mm

Changes to the substrate core thickness provided little to no change in the thermal properties of the package as seen in Table 9. It can be seen that the substrate core thickness of 0.41mm provided a slight improvement in lowering the maximum temperature of the package but not by much when comparing to the thicknesses of 0.5mm and 0.6mm.

Table 9. Thermal performance for different substrate core thickness

	Substrate Core Thickness		
	0.41mm	0.5mm	0.6mm
Maximum temperature, °C	48.874	48.893	48.915
Convection heat dissipation path on the lid, W	-9.7613	-9.7683	-9.7764
Convection heat dissipation path on the substrate, W	-0.23873	-0.23168	-0.22355
Radiation heat dissipation on the lid, W	-1.1374e-007	-1.1383e-007	-1.1394e-007

Analysis of the temperature towards the bottom and top of the package as seen in Fig. 14 and Fig. 15 shows that all three substrate core thicknesses perform very similarly. This lack of difference is due to the fact that the package is designed to remove heat towards the top with the TIM interfacing with the lid which helps remove heat from the package.

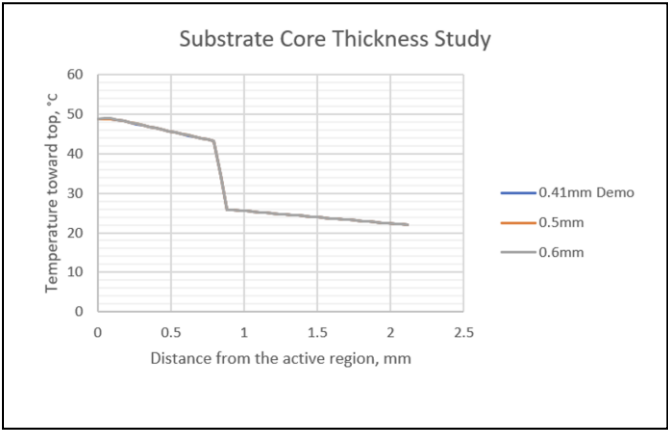


Figure 14. Temperature toward top of the package (Substrate Core Thickness)

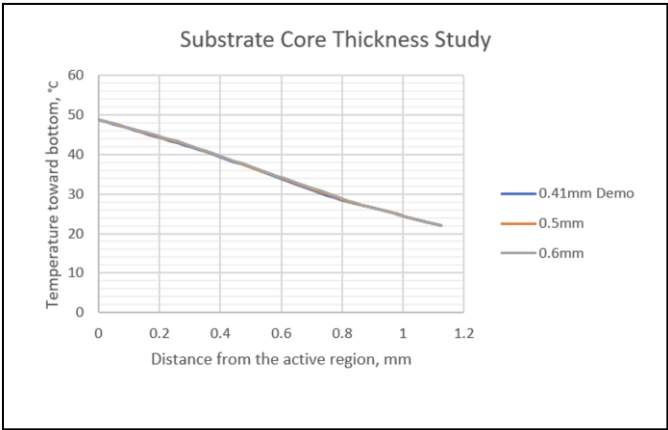


Figure 15. Temperature toward bottom of the package (Substrate Core Thickness)

D. Bulk Silicon Thickness: 0.7mm vs 0.5mm vs 0.3mm

The final thermal analysis that was performed on the package was changing the thickness of the bulk silicon. As seen in Table 10, the thinnest bulk silicon with a thickness of 0.3mm performed the best by having the lowest maximum temperature of 45.797 °C.

Table 10. Thermal performance for different bulk silicon thickness

	Bulk Silicon Thickness		
	0.7mm	0.5mm	0.3mm
Maximum temperature, °C	48.874	47.314	45.797
Convection heat dissipation path on the lid, W	-9.7613	-9.7759	-9.7907
Convection heat dissipation path on the substrate, W	-0.23873	-0.22409	-0.20931
Radiation heat dissipation on the lid, W	-1.1374e-007	-1.1393e-007	-1.1411e-007

The data presented in Fig. 15 and Fig. 16 confirms that the thinnest bulk silicon trial with a thickness of 0.3mm performs the best by having a lower temperature through the package towards the top and bottom.

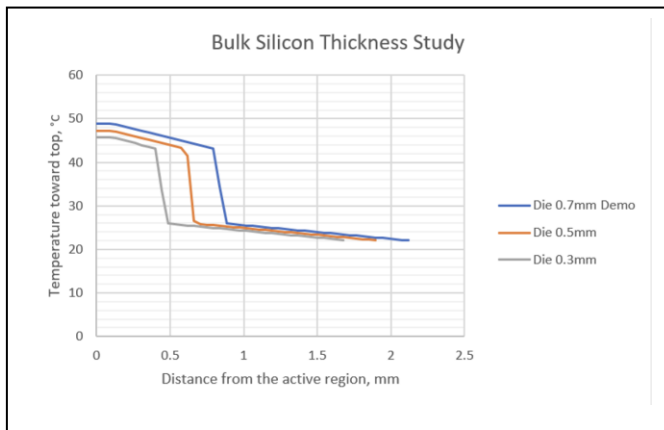


Figure 15. Temperature toward top of the package (Bulk Silicon Thickness)

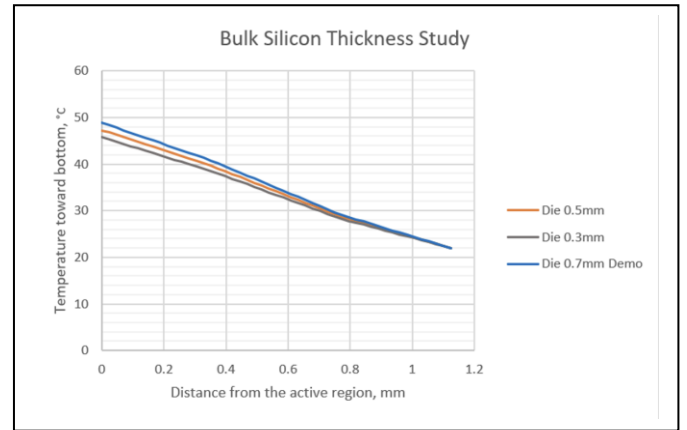


Figure 16. Temperature toward bottom of the package (Bulk Silicon Thickness)

VII. CONCLUSION

Through the completion of the parametric studies outlined in this report, comparisons of thermal performance between different material properties and geometries were explored. It was shown that changing these properties had a significant impact on the thermal performance of the flip chip package. These changes in thermal performance between the different designs will have an impact on the reliability of the package by exerting mechanical stress through temperature cycling. By studying the effect of these material properties on the thermal performance of the package, the possible mechanical failures can be understood better.

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